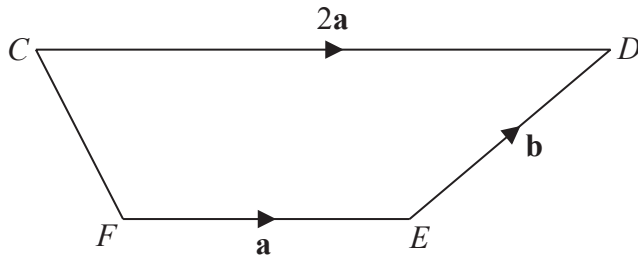


**24**  $CDEF$  is a quadrilateral.



$$\vec{FE} = \mathbf{a} \quad \vec{ED} = \mathbf{b} \quad \vec{CD} = 2\mathbf{a}$$

The point  $P$  is such that  $CEP$  is a straight line and that  $CE = EP$

Use a vector method to prove that  $CF$  is parallel to  $DP$ .

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**(Total for Question 24 is 4 marks)**